

AYUSH PALLOD

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EDUCATION

Timeline	Degree & Specialization	Institute	CGPA
2025 – 2027	M.Tech in IT (Machine Learning & Intelligent Systems)	Indian Institute of Information Technology (IIIT), Allahabad	9.43
2021 – 2025	B.E in Computer Science and Engineering	Osmania University, Matrusri Engineering College, Hyderabad	8.72

SKILLS AND TOOLS

- **Programming Languages:** Python, C, C++, SQL
- **ML / DL & Agentic Frameworks:** PyTorch, TensorFlow, Keras, Hugging Face (Transformers), LangChain, LangGraph
- **Data Libraries:** Scikit-learn, NumPy, Pandas, OpenCV, NLTK, Pillow, Matplotlib, Streamlit, SQLAlchemy
- **MLOps & Cloud Infrastructure:** Linux, SSH, Git, Docker, NVIDIA T4 & L4 GPUs, E2E Networks, Kaggle
- **Experiment Tracking & Serving:** Weights & Biases (W&B), Ollama, MLflow, Evidently AI

EXPERIENCE

- Amptronics Systems Pvt. Ltd.** | Data Analyst Intern **May 2024 – June 2024**
- Engineered **real-time data analysis pipelines** for the Wireless Bus Bar Temperature Management System (WB-TMS), reducing operational latency through dynamic HTML-based reporting dashboards.
 - Optimized **relational query structures** across Azure Database and SQLite architectures, establishing reliable storage mechanisms for core application telemetry.
 - Automated **end-to-end operational workflows** based on target client requirements, eliminating manual intervention overheads by roughly 35%.
- Slash Mark** | Data Analyst Intern **Nov 2023 – Feb 2024**
- Analyzed and visualized large-scale, multi-variate datasets utilizing Pandas, Matplotlib, and Seaborn to isolate critical trends and deliver **actionable asset insights**.

PROJECTS

- FinSight AI – Enterprise Financial Intelligence Platform** 🌐 | FastAPI, Qdrant, PostgreSQL, Redis, Celery, Ollama, Docker, Prometheus, Grafana **2026**
- Built a **production-grade hybrid RAG platform** combining **BM25**, **BGE-M3 dense retrieval**, and **RRF** to deliver cited answers from **10-K**, **10-Q**, and earnings transcript documents.
 - Engineered a **9-service Docker Compose architecture** (FastAPI, PostgreSQL, Redis, Celery, Qdrant, Ollama, Prometheus, Grafana, Nginx) with async ingestion/OCR/embedding pipelines, evaluated via **RAGAS** (faithfulness, context precision/recall, answer relevancy).
 - Implemented **prompt-injection detection** and **PII redaction** guardrails; instrumented **Prometheus/Grafana** observability for p50/p95/p99 latency, retrieval and generation latency, with automated **GitHub Actions CI/CD** deployment.
- AI Multi-Agent Stock Analyst Bot** 🌐 | LangChain, RAG, Ollama, Streamlit, Financial Analysis **2026**
- Architected a **LangChain-based multi-agent system** to orchestrate technical, sentiment, and fundamental stock analysis, enabling modular, production-style financial reasoning workflows.
 - Deployed a **self fine-tuned LLaMA 3.2 domain-specific LLM** alongside **LLaMA 3.1 (7B)** and **Phi-3 Mini** to maximize financial sentiment extraction accuracy and model execution consistency.
 - Implemented **MLOps reliability frameworks** including a custom **RAG pipeline** for grounded execution, automated out-of-scope query rejection layers, and **Ollama-based inference serving**.
- Stochastic Finance & Valuation Platform** 🌐 🌐 | Numba JIT, ARIMA, Monte Carlo, GitHub Actions **2026**
- Built a high-throughput quantitative engine running **Monte Carlo asset evaluations** and custom **ARIMA-driven ensemble forecasting** architectures.
 - Realized a **50× execution speed improvement** via **Numba JIT acceleration** of nested iterative pathways, integrating multi-asset tracking matrix features.
 - Deployed **live on Streamlit Cloud** with a **GitHub Actions CI pipeline** validating GBM, JIT equivalence & DCF logic.

CERTIFICATIONS & SCHOLASTIC ACHIEVEMENTS

- **Winner**, OOC Hackathon 2026 – Organised by Indian Institute of Information Technology (IIITA). **2026**
- **Ranked Top 2% Nationwide**, GATE (CS/IT) – Obtained a normalized examination score of 593. **2025**
- **Innovative Project Award**, CSI Hyderabad Chapter – For creating a neural-driven Brain Tumor detection framework. **2024**
- **Machine Learning Specialization** – Guided by Andrew Ng via Coursera (Stanford University & DeepLearning.AI).
- **MLOps Zoomcamp Certification** – Developed practical engineering competencies through DataTalks.Club curricula.